well as by its inertia. The thrust prevents the frame's motion from being uniform. A reference frame attached to the space shuttle as it reenters the Earth's atmosphere also is not inertial, because friction between the shuttle's skin and the Earth's air molecules slows the shuttle, making its motion nonuniform.

Most important, near any massive body such as the Earth, *all* reference frames are pulled by gravity. There is no way whatsoever to shield a reference frame (or any other object) from gravity's pull. Therefore, by restricting himself to inertial frames, Einstein prevented himself from considering, in 1905, physical situations in which gravity is important[3];in effect, he idealized our Universe as one in which there is no gravity at all. Extreme idealizations like this are central to progress in physics; one throws away, conceptually, aspects of the Universe that are difficult to deal with, and only after gaining intellectual control over the remaining, easier aspects does one return to the harder ones. Einstein gained intellectual control over an idealized universe without gravity in 1905. He then turned to the harder task of understanding the nature of space and time in our real, gravity-endowed Universe, a task that eventually would force him to conclude that gravity warps space and time (Chapter 2).

With the concept of an inertial reference frame understood, we are now ready for a deeper, more precise formulation of Einstein's principle of relativity: *Formulate any law of physics in terms of measurements made in one inertial reference frame. Then, when restated in terms of measurements in any other inertial frame, that law of physics must take on precisely the same mathematical and logical form as in the original frame.* In other words, the laws of physics must not provide any means to distinguish one inertial reference frame (one state of uniform motion) from any other.

Two examples of physical laws will make this more clear:

- "Any free object (one on which no forces act) that initially is at rest in an inertial reference frame will always remain at rest; and any free object that initially is moving through an inertial reference frame will continue forever forward, along a straight line with constant speed." If (as is the case) we have strong reason to believe that this relativistic version of Newton's first law of motion is true in at least one inertial reference frame, then the principle of relativity insists that it must be true in all inertial reference frames regardless of